Bernoulli-EKF scene-graph tracker: per-frame pipeline

1 Scope

Implementation map for one frame of `TwoTierOrchestrator.step()` in `pose_update/orchestrator.py`, with `B-eq:*` cross-references into `bernoulli_ekf.tex` (the formulation paper, authoritative on ambiguity). Per-track state lives in *base frame*: $X^{(i)} = (\ell, T_{bo}, P_{bo}, r, \tau, \mathcal{H})$. SLAM covariance Σ_{wb} never enters the recursion (it appears only in the world-frame emit).

2 Pipeline overview

Standard MOT order, predict-then-measure:

$$
\begin{aligned}
&(T_{wb,k}, T_{bc}(t_k), T_{bg,k},\ \mathrm{phase}_k,\ \{m_l, \hat{\ell}_l, \hat{s}_l, \tau_l\}_l) \\
&\quad \xrightarrow{\textbf{0/1}\ \text{ingest+predict}} (T,P)_{k|k-1} \xrightarrow{\textbf{2}\ \text{centroid (cleaned mask)}} \{c_l^{co}\} \\
&\quad \xrightarrow{\textbf{3}\ \text{Hungarian on } d_t^2} \pi^* \xrightarrow{\textbf{3b}\ \text{ICP at filter prior}} \{(\hat{T}_l^{co}, \hat{R}_l^{co})\}_{l\in\pi^*} \\
&\quad \xrightarrow{\textbf{4-6}\ \text{vis+update+miss}} X_k \xrightarrow{\textbf{7}\ \text{pending+admit}} X_k \cup X^{\text{new}} \xrightarrow{\textbf{8}\ \text{prune+merge}} \widehat{\mathcal{X}}_k
\end{aligned}
$$

Two non-trivial design constraints.

- **Tracker oid is state-only.** The detection's `id` (perception's per-mask tracking id, e.g. SAM2's `tau`) is metadata, never tracker identity. Tracker oids are minted by `_next_track_id()` (fresh consecutive int) at admission only.
- **ICP is a measurement, not an admission gate.** For matched pairs ICP refines (T, R); for births ICP runs *exactly once on admission* (centroid-anchored, populates `PoseEstimator._refs[oid]`). The pre-2026 "ICP probes every unmatched detection then admit" path leaked stale reference clouds and is removed.

Per-frame algorithm (one line each).

0. Ingest $T_{wb,k}$, install $T_{bc}(t_k)$ (§3.1); cache $T_{bg,k}$ for held tracks.

1. Predict: static tracks via $u_k = T_{wb,k}^{-1} T_{wb,k-1}$; held tracks via ΔT_{bg} with phase-dependent Q (§4); $r \leftarrow p_s r$.

2. Per-mask cleaned-cloud centroid $c_l^{co} \in \mathbb{R}^3$.

3. Hungarian on the 3-DOF d_t^2, $\chi_3^2(0.9997) = 21.108$ gate.

3b.
One ICP per matched pair, seeded with $\hat{T}_{k|k-1,i}^{co} = T_{cb}(t_k) T_{bo,k|k-1}^{(i)}$ against `_refs[oid]`.

4. Visibility $p_v^{(i)}$.

5. Matched update: IEKF + Joseph + Huber on d_t^2; r via `B-eq:r_assoc`; chain append.
6. Missed: r via `B-eq:r_miss`; T, P unchanged.
7. **Birth via pending buffer**: policy gate (border + tracker-side $n_{\text{obs}} \geq K$ + score) $\rightarrow$ if admitted, `_next_track_id()` + ICP once + commit (§3.9).
8. Prune $r < r_{\min}$; greedy self-merge with held-id-as-keeper (§4).
9. Emit $\{i : r \geq r_{\text{conf}}\}$; world-frame composition $T_{wo} = T_{wb} T_{bo}$.

3 Per-frame steps

3.1 Step 0. Ingest + per-frame camera extrinsic

`ingest_slam` caches $T_{wb,k}, T_{wb,k-1}$ for the predict's body-form control $u_k = T_{wb,k}^{-1} T_{wb,k-1}$. The world-form alternative $T_{wb,k-1} T_{wb,k}^{-1}$ produces $\sim$1 m drift on real trajectories (adjoint-conjugation mismatch).

The base-to-camera-optical extrinsic $T_{bc}(t_k)$ is *time-varying* on Fetch (head pan/tilt + torso lift). The driver loads it from `pose_txt/T_bc.txt` (one row per frame) and installs it via `set_camera_extrinsic` *before* any measurement-side call. Treating T_{bc} as constant aliases head motion as object motion and silently inflates d_t^2 on every panned frame. Static-predict u_k is T_{bc}-free; only the measurement lift and the ICP prior depend on it.

3.2 Step 1. Predict (state + existence)

Static tracks (the default, `predict_static` with `skip_oids=held_oids`): mean $\mu \leftarrow u_k \mu$, $P \leftarrow \Phi(\mathrm{Ad}(u_k) P \, \mathrm{Ad}(u_k)^\top + Q)$, then per-axis floor (`cfg.P_min_diag`, defaults 5 mm/0.05 rad). The floor is not cosmetic: without it, P shrinks below per-frame perception jitter after $N \gg 1$ updates and the χ_3^2 gate spuriously rejects 2–3 cm same-object drift.

Held tracks (separate branch, `rigid_attachment_predict`): $\mu \leftarrow \Delta T_{bg} \mu$ with $\Delta T_{bg} = T_{bg,k} T_{bg,k-1}^{-1}$, $P \leftarrow \mathrm{Ad}(\Delta T_{bg}) P \, \mathrm{Ad}(\Delta T_{bg})^\top + Q_{\text{manip}}$. Held tracks must be *skipped* from `predict_static`; otherwise the static u_k composes onto $\Delta T_{bg} \mu$ and the kinematics $T_{bo} = T_{bg} T_{go}$ break.

Q is selected by `process_noise_for_phase(phase, is_target, frame, frames_since_obs)`:

- *Static* ($\neg$is_target): Q decays with `frames_since_obs` from $\mathcal{N}(0, 10^{-6})$ down to $\mathcal{N}(0, 10^{-8})$ per axis.
- *Held, holding*: $Q = \mathrm{diag}(2.5 \cdot 10^{-4}, 10^{-3})$ per trans / rot axis. Models grip slack + proprioception jitter.
- *Held, grasping/releasing*: $Q = I$ (essentially unknown). The transition frame's pose is not a function of ΔT_{bg}.

Bernoulli predict: $r_{k|k-1} = p_s r_{k-1|k-1}$ (default $p_s = 1$).

3.3 Step 2. Coarse measurement (cleaned mask centroid)

For each detection: $c_l^{co} =$ `centroid_cam_from_mask`(m_l, D_k, K). The mask is first passed through `_clean_mask` (`pose_update/icp_pose.py`):

1. Binary erosion (1 pixel) drops the worst boundary band.
2. Per-masked-pixel 4-neighbour depth-jump rejection ($|\Delta d| > 3$ cm kills boundary pixels straddling foreground/background; object curvature is sub-cm).

Both stages *self-disable* below 30 surviving pixels (small masks keep their thin rim). On `apple_in_the_tray` frame 300 (a $\sim$15k-pixel tray detection): retains 90% of pixels, shrinks `obj_radius` from 19.9 to 16.4 cm by removing background leakage tails. Centroid shift is sub-mm (filter is spatially symmetric); ICP fitness stable.

The Hungarian cost uses isotropic $R_{\text{cent}} = \text{diag}((2\,\text{cm})^2 I_3)$. Detections that fail back-projection (<30 valid-depth pixels even after cleaning) are dropped from association *and* from the birth buffer.

3.4 Step 3. Hungarian association

`association.hungarian_associate`, called with a 3-DOF centroid innovation:

$$\nu = [\nu_{3D}, \mathbf{0}]^\top \in \mathbb{R}^6, \quad S = \text{diag}(S_{3D},\, \tau I_3), \tag{1}$$

$$d_t^2 = \nu_{3D}^\top S_{3D}^{-1} \nu_{3D}, \quad d_r^2 = 0 \tag{2}$$

with $\tau = 10^6$ in the rotation slot to make the block decomposition cleanly 3-DOF. $S_{3D} = P_{tt}^{(i)} + R_{\text{cent}}^{bo}$ where $R_{\text{cent}}^{bo} = R_{bc} R_{\text{cent}}^{cam} R_{bc}^\top$ is the rotation-only base-frame lift of the camera centroid covariance. A naive 6-DOF $R_{icp} = \text{diag}(R_t, \tau I)$ would couple τ into translation via $[t_{bc}]_\times$ and collapse $d_t^2 \to 0$.

Cost (in χ^2 units, `cost_d2_mode='sum'`):

$$C_{il} = (d_{il,t}^2 + 0.1\, d_{il,r}^2) \;-\; \alpha\, \mathbb{1}[\tau_l = \tau^{(i)}, \tau^{(i)} \geq 0] \;+\; \beta\, \mathbb{1}[\hat{\ell}_l \notin \mathcal{H}^{(i)}] \;+\; \gamma \max(0, 1 - \hat{s}_l) \tag{3}$$

with $\alpha = 4.4$ (SAM2-id continuity), $\beta = 6.0$ (label-flicker soft penalty, $< G_{\text{out},t}$ so geometry can override label), $\gamma = 2.0$ (low-score mild penalty). Outer gate $d_{il,t}^2 \leq 21.108$ (`gate_mode='trans'`); cells failing it get the sentinel `_INFEASIBLE`$=10^{12}$.

Solver: `scipy.optimize.linear_sum_assignment` on the rectangular cost matrix; assignments at `_INFEASIBLE` are post-filtered out (Hungarian on a fully-infeasible row will still emit some column). Result: `match{oid: det_idx}`.

Recovering the paper baseline: `enforce_label_match=True`, $\beta = \gamma = 0$, `gate_mode='full'`, `cost_d2_mode='full'`. **Oracle mode**: `association_mode="oracle"` bypasses the cost matrix and matches by upstream `det["id"]`; used by the degeneracy test.

3.5 Step 3b. Per-match fine ICP

For every $(i, l) \in \pi^*$: `PoseEstimator.estimate(oid=i, mask=m_l, depth=D, T_co_init=` $\hat{T}_{k|k-1,i}^{co}$`)`. The seed is the matched track's filter prior; ICP refines against `_refs[i]` (*tracker-oid*-keyed, stable across SAM2 id reissues for the same physical object).

Output $\hat{R}^{co}$ uses data-driven floors:

$$\sigma_t^2 = \max(\text{rmse}^2, (2\,\text{cm})^2) \cdot \text{fit_scale},$$
$$\sigma_r^2 = \max((\text{rmse}/r_{\text{obj}})^2, (0.1\,\text{rad})^2) \cdot \text{fit_scale}.$$

ICP failure (`fitness` < 0.90 or `rmse` > 15 mm) demotes the pair to a centroid-only update with inflated rotation R (no rotation information through Kalman).

3.6 Step 4. Visibility $p_v^{(i)}$

$p_v^{(i)} = \mathbb{1}[\pi_K(T_{bc}^{-1} T) \in \Omega] \cdot \prod_{j \neq i}(1 - \rho_{ij})$ (§9). Implementation: `visibility.visibility_p_v`. Off when `enable_visibility=False`; Production orchestrator still uses constant T_{bc} here (test driver uses per-frame).

3.7 Step 5. Matched-track update

For matched (i, l): lift the camera-frame measurement to base via $T_{bo,\text{meas}} = T_{bc}(t_k)\hat{T}^{co}$, $R^{bo} = \mathrm{Ad}(T_{bc})\,\hat{R}^{co}\,\mathrm{Ad}(T_{bc})^{\top}$. Then IEKF (2 inner iterations) + Joseph form + Φ saturate + diagonal floor on P.

Existence update: $r \leftarrow$ `r_assoc_update_loglik`$(r, \log\mathcal{L}, p_d, \lambda_c)$ in log-space (avoids underflow for $\log\mathcal{L} \sim -50$).

Robust gate: Huber weight on d_t^2 ($G_{\text{in},t}$ inner soft, $G_{\text{out},t}$ outer hard); $w = 0$ tracks revert to the miss branch. Append $(k, \hat{T}^{co}, \hat{R}^{co})$ to the loop-closure-aware chain $\mathcal{H}^{(i)}$ (§11).

3.8 Step 6. Missed-track update

For unmatched i: $r \leftarrow (1-\tilde{p}_d)r/((1-\tilde{p}_d)r + (1-r))$, $\tilde{p}_d = p_d p_v^{(i)}$. T, P unchanged. When $p_v \to 0$ the update collapses to $r_{k|k} = r_{k|k-1}$.

3.9 Step 7. Birth via pending buffer

Two-phase, with the *tracker* owning admission counters:

Pending buffer (`self._pending_births`, dict keyed by perception id). For each unmatched detection l that passes the back-projection gate:

1. Look up / create `_PendingBirth(source_id=det["id"], ...)`; bump `n_obs_tracker`, update `max_score`.
2. **Policy gate** (`birth_admissible` with `require_pose=False`): bbox not on image border ($\geq$2px), `n_obs_tracker`$\geq K = 3$, `max_score`$\geq S = 0.20$. Note: K counts the *tracker*'s unmatched-streak frames, not perception's `n_obs` (which resets on SAM2 id reissue).
3. If admitted: mint i_{new} = `_next_track_id()` (fresh consecutive int, NOT seeded from `det["id"]`); call `PoseEstimator.estimate(oid=i_new, T_co_init=None)` once (first-observation branch: anchors at c_l^{co}, populates `_refs[i_new]`); commit via `_birth_track`; pop the pending entry. Bernoulli init: $r = \lambda_b\hat{s}/(\lambda_b\hat{s} + \lambda_c)$.
4. If rejected: keep the pending entry. TTL-expire after `birth_pending_ttl_frames`=30 of no re-arrival (one second at 30 Hz; survives short perception dropouts).

This invariant matters: *$i \in$ `_refs` iff i is a committed track.* The pre-2026 path called `estimate` as an admission probe, seeding `_refs[i]` before the policy gate and leaking it on rejection; subsequent frames then ran refine-ICP against that stale cloud and silently dropped real birth candidates (manifested as the apple-in-tray + tray failing to ever birth).

Birth covariance: $P_{\text{new}} = \mathrm{diag}((5\,\text{cm})^2, (0.05\,\text{rad})^2)$ (default; `init_cov_from_R=True` uses the ICP $\hat{R}$ instead).

3.10 Step 8. Prune + self-merge

Prune (`_prune_track`): drop i with $r < r_{\min} = 10^{-3}$ from `state.objects`, all per-orchestrator dicts, and `_refs[i]`.

Self-merge (`_self_merge_pass(held_id)`): Hungarian's one-to-one constraint leaves duplicates when SAM2 emits parallel ids for one physical object. Per-frame greedy fusion of same-label pairs with $\|\mu_i - \mu_j\| \leq$ `self_merge_trans_m` $= 5\,\text{cm}$. Keep decision: held oid wins (preserves rigid-attachment kinematics); otherwise longer chain wins; tie-break lower oid. Fusion is Bayesian information sum on SE(3) linearised at μ_{keep}:

$$P_{\text{new}}^{-1} = P_{\text{keep}}^{-1} + P_{\text{drop}}^{-1}, \quad \mu_{\text{new}} = \mu_{\text{keep}}\,\mathrm{Exp}\big(P_{\text{new}}P_{\text{drop}}^{-1}\,\mathrm{Log}(\mu_{\text{keep}}^{-1}\mu_{\text{drop}})\big). \tag{4}$$

Bookkeeping: $r \leftarrow$ max, `frames_since_obs`$\leftarrow$ min, `label_scores` per-label sum + sample-weighted mean, $\mathcal{H}^{(\text{drop})}$ appended to $\mathcal{H}^{(\text{keep})}$, `_refs[drop]` merged into `_refs[keep]` via voxel-downsample.

3.11 Step 9. Emit

`TwoTierOrchestrator.objects` returns $\{i : r^{(i)} \geq r_{\text{conf}} = 0.5\}$. Tentative tracks are still predicted/associated/updated but hidden; visible via `tentative_objects`. World-frame composition $T_{wo} = T_{wb} T_{bo}$ (or the chain smoother of §11 when bundle-adjusted).

4 Held-object handling

The dataset has no annotated `phase`; the test driver's `_GripperStateInferrer` (`tests/visualize_ekf_tracking.py`) infers $\{\text{phase}, \text{held_obj_id}\}$ from proprioception:

Phase from gripper finger width. $w = \text{l_gripper_finger_joint} + \text{r_gripper_finger_joint}$ (from `joints_pose.json`). Hysteresis bands prevent chatter:

$$\text{is_closed}_k = \begin{cases} w_k < 0.040\,\text{m}, & \text{previously closed}, \\ w_k < 0.025\,\text{m}, & \text{previously open}. \end{cases}$$

Phase transitions: $\{\text{open} \rightarrow \text{closed}\} = \textit{grasping}$, $\{\text{closed} \rightarrow \text{closed}\} = \textit{holding}$, $\{\text{closed} \rightarrow \text{open}\} = \textit{releasing}$, otherwise *idle*.

Held identity from EE proximity. At grasp onset (idle$\rightarrow$grasping) compute the world-frame EE position $T_{we} = T_{wb} T_{bg}$ and pick the nearest live track within `grasp_radius_m`=30 cm; that becomes `held_obj_id`. The id is *frozen* through holding (no per-frame re-inference – that would jitter as masks resegment), released at the *releasing* edge, and migrated through self-merges via `apply_merges` (which reassigns to the keeper if the held oid was a drop).

Q schedule. The orchestrator passes $\{T_{bg}, \text{held_oids}, \text{phase}\}$ to `tracker.step()`. `rigid_attachment_predict` reads the phase-dependent Q from `process_noise_for_phase(phase, is_target=True, frame="base")`: huge during grasping/releasing (transition pose is unknown), $\sim$1.6 cm/1.8° per frame during holding (grip slack + proprio jitter). The pre-fix $\mathcal{N}(0, 10^{-8})$ value was unrealistically tight; P stayed pinned at the floor and the Hungarian gate rejected every re-detection of the moving held object.

Self-merge protection. `_self_merge_pass(held_id=...)` forces the held oid as `keep` in any pair it participates in, preserving the rigid-attachment motion model across merges.

5 Code map (quick reference)

Step	Equation / role	Code pointer
0	$T_{bc}(t_k)$ install	`GaussianState.set_camera_extrinsic`
1	Exp / Log, predict, Φ	`ekf_se3.{se3_exp,se3_log,saturate_covariance}`
1	static + held predict	`predict_static`, `rigid_attachment_predict`
1	Q schedule	`ekf_se3.process_noise_for_phase`
1	$r_{k\|k-1} = p_s r$ (`B-eq:bern_pred_r`)	`bernoulli.r_predict`
2	cleaned mask + centroid	`icp_pose.{_clean_mask,centroid_cam_from_mask}`
3	cost + Hungarian (`B-eq:cost,B-eq:hung`)	`association.hungarian_associate`
3b	per-match ICP	`PoseEstimator.estimate` (refine branch)
4	p_v (`B-eq:pv,B-eq:rho`)	`visibility.visibility_p_v`
5	innovation, IEKF update, Joseph, Huber	`GaussianState.update_observation`, `ekf_se3.huber_wei`
5	r (`B-eq:r_assoc`)	`bernoulli.r_assoc_update_loglik`
6	r (`B-eq:r_miss`)	`bernoulli.r_miss_update`
7	policy gate	`birth_admissible` (`require_pose=False`)
7	pending buffer	`_PendingBirth`, `_pending_births` dict
7	admit + commit (`B-eq:birth_r,B-eq:birth_state`)	`_next_track_id` + `_birth_track`
7	ICP at admit (first-obs branch)	`PoseEstimator.estimate(T_co_init=None)`
8	prune	`_prune_track`
8	self-merge (`B-eq:self_merge_info`)	`_self_merge_pass(held_id=...)`
9	emit $r \geq r_{\text{conf}}$	`TwoTierOrchestrator.objects`
held	phase + held oid inference	`_GripperStateInferrer`

6 Known deviations from `bernoulli_ekf.tex`

1. Standalone `ekf_se3.ekf_update` uses $(I - K)P$ rather than Joseph and skips Φ. The RBPF Bernoulli path (`RBPFState.update_observation`) is paper-correct.
2. Production orchestrator still passes constant T_{bc} to the visibility predicate; the test driver installs per-frame $T_{bc}(t_k)$ via `set_camera_extrinsic`. Migrate by adding the call in `step()` before any measurement-side function; until then, set `enable_visibility=False` on the production path or accept head-pan-aliased visibility flicker.
3. Visibility footprint $A^{(i)}$ uses the matched-detection bbox as a proxy. Paper prescribes projecting the TSDF shape summary $\mathcal{G}^{(i)}$.
4. *Mask-edge cloud filter* (§3.3) and *rigid-attachment predict with phase-dependent Q* (§4) are manipulation-aware extensions not in the paper.
5. *Pending-birth buffer* (§3.9) replaces the paper's per-frame "unmatched detection → immediate birth" with a tracker-side admission policy. The Bernoulli existence r semantics are unchanged; only the gating before r_{new} is computed differs.